X(3872) Analysis Update

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What changed since September 24?

- **2018 added:** separate ML for each year; combined fit for 2018 and 2023.
- **More variables with physical meanings:** leading/subleading pion p_T; maximum/minimum $\Delta R(\pi, J/\psi)$; maximum pion $|\eta|$; leading muon p_T and maximum $\Delta R(\mu, J/\psi)$.
- **MC reweighting corrected:** remove the pthat 5 component; recompute weights in a correct way.

Current MC reweighting

pthat component threshold t_j; generated $p = \hat{p}_T$; generated count N_j, cross section σ_j, filter efficiency ϵ_j.

2023: production normalization table

$$L_j = \frac{N_j}{\sigma_j \epsilon_j}, \qquad w(p) = \left(\sum_{\substack{j \in \{10,15,30,50\} \\ t_j \le p}} L_j \right)^{-1}.$$

N_j: full generated count, before selection.

2018: infer retained contributions

$$w_{\text{new}} = \frac{1}{1/w_{\text{old}} - 1/w_5}.$$

w_5: old lowest-bin weight. Assumes the historical cumulative normalization; preserves its overall scale.

Zero weight: pthat 5 component, or $p < t_j$ for component j. Retained intervals: $[10, 15)$, $[15, 30)$, $[30, 50)$, $[50, \infty)$ GeV.

ML uses `pThatreweight`> 0; no additional data-driven reweighting.

Why exclude the pthat 5 component?

Original prompt X MC: $15 < \texttt{Bpt} < 50$, $\mathrm{abs}(\texttt{By}) < 1.6$, $\texttt{BQvalue} < 0.15$

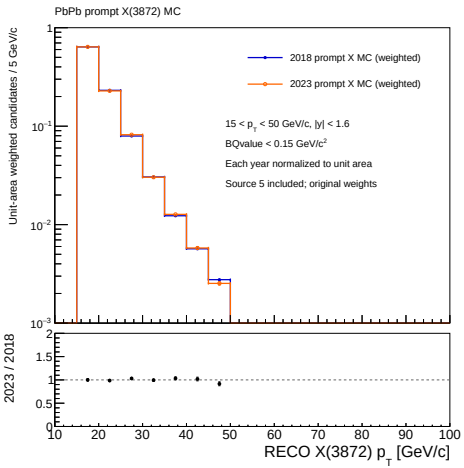
	2018	2023
pthat 5 component: candidates, all generated $\hat{p}_T$	28,188	851
pthat 5 component: fraction of reweighted entries	44.95%	23.18%
pthat5 component, $\hat{p}_T < 10$: positive-weight count	5,994	164
pthat5 component, $\hat{p}_T < 10$: fraction of reweighted entries	19.83%	19.62%

Motivation: reduce the influence of very few, highly weighted 2023 candidates.

Open question: why do 2018 and 2023 differ so much in the pthat 5 component? The generation/filter/reconstruction chain is not fully reconciled.

Removing the pthat 5 component also removes its high-$\hat{p}_T$ tail; weights must be recomputed.

Reweighted prompt X MC in the fiducial region



$15 < \texttt{Bpt} < 50$ GeV
$|\texttt{By}| < 1.6$
$\texttt{BQvalue} < 0.15$ GeV

pthat 5 component included.

Each year normalized to unit area after selection.

Although the pthat 5 component behaves differently in 2018 and 2023, the p_T distributions after reweighting are very similar.

Common selection and yearly differences

Cut before ML

Variable	Requirement
`Bpt`	$15 < p_T < 50$ GeV
`By`	$\lvert y \rvert < 1.6$
`BQvalue`	$Q < 0.15$ GeV

Signal: prompt X MC; positive MC weights.

Background: DATA sidebands $(3.80, 3.85) \cup (3.89, 3.94)$ GeV; unit weights.

Yearly difference

2018: `Bchi2Prob`> 0.1 already imposed upstream.

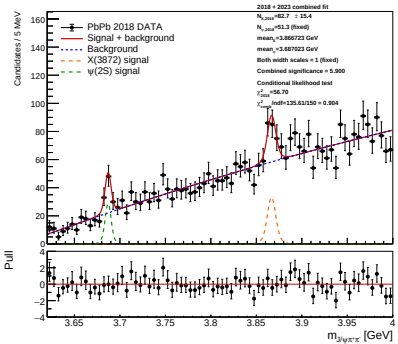
2023: no additional `Bchi2Prob`> 0.1 requirement in the models shown here.

Cut after ML

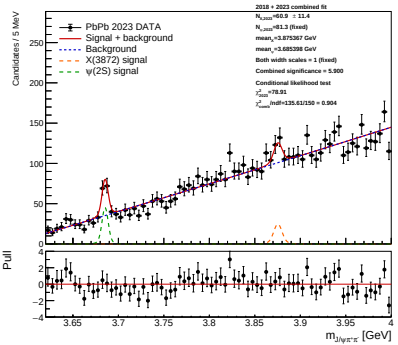
Each model uses its own Punzi-optimal score, selected with the training sideband range and the total MC sample.

Combined fit: Btrk1dR, Btrk2Pt, Btrk1Pt

2018 score > 0.86



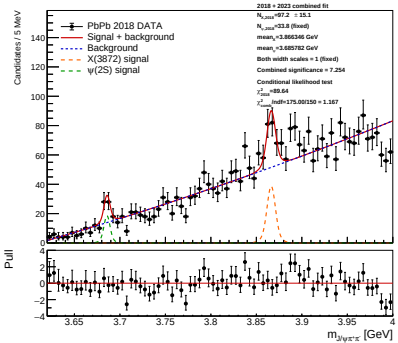
2023 score > 0.84



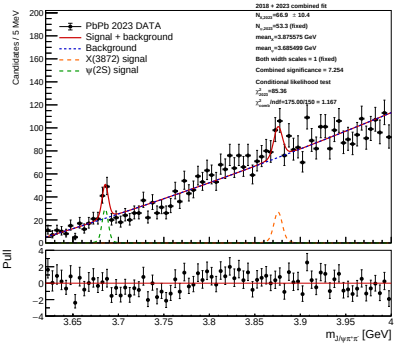
Combined: $N_X = 143.60 \pm 26.74$; significance $= 5.900\sigma$.
Independent-year reference: 2018: 5.066σ; 2023: 2.958σ.
Combined: $\chi^2/\mathrm{ndf} = 0.904$.
MC double-Gaussian widths fixed; 2nd-order Chebyshev background.

Combined fit: BtrkLeadPt, BtrkSubPt, BtrkMaxdR

2018 score > 0.86



2023 score > 0.86



Combined: $N_X = 164.10 \pm 25.45$; significance $= 7.254\sigma$.

Independent-year reference: 2018: 6.188σ; 2023: 3.722σ.

Combined: $\chi^2/\mathrm{ndf} = 1.167$.

MC double-Gaussian widths fixed; 2nd-order Chebyshev background.

Current preferred candidates

Variable set	3v3	3v7	9v1	5v1	3v5	3v12
Combined significance [σ]	5.900	7.254	4.703	4.410	5.573	5.961

- 3v3: `Btrk1dR`, `Btrk2Pt`, `Btrk1Pt`.
- 3v7: `BtrkLeadPt`, `BtrkSubPt`, `BtrkMaxdR`.
- 9v1: `Bmu1pt`, `Bmu1eta`, `Bmu2pt`, `Bmu2eta`, `Btrk2Pt`, `BtrkPtimb`, `Btrk1dR`, `Bujeta`, `Btktketa`.
- 5v1: `Bmu1pt`, `Bmu2pt`, `Btrk2Pt`, `BtrkPtimb`, `Btrk1dR`.
- 3v5: `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`.
- 3v12: `BtrkLeadPt`, `BtrkPtimb`, `BtrkMaxdR`.

- **Possible explanation:** for a rare X signal, more variables may make high-dimensional MC mismodelling or overfitting more pronounced.
- **Current preferred candidates:**

  `Btrk1dR`, `Btrk2Pt`, `Btrk1Pt`

  and

  `BtrkLeadPt`, `BtrkSubPt`, `BtrkMaxdR`.

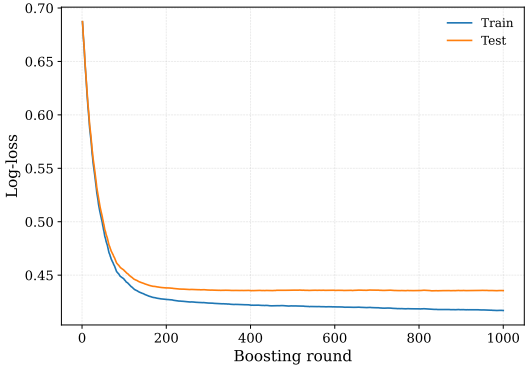
Thanks for listening!

Reporter: Lekai Yao

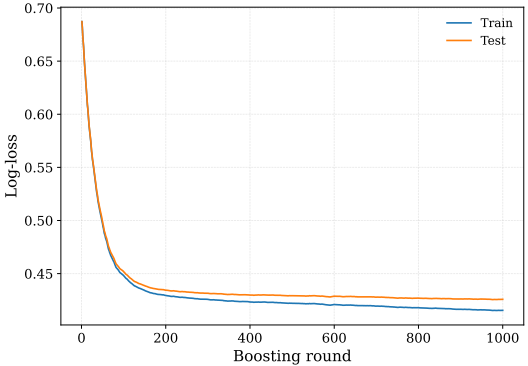
2026.9.30

Logloss: Btrk1dR, Btrk2Pt, Btrk1Pt

2018



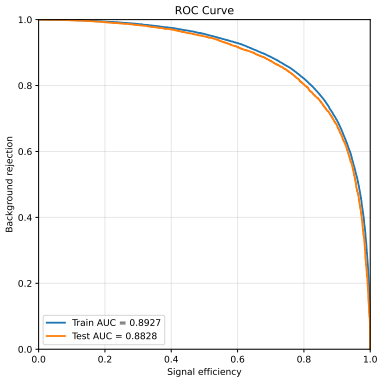
2023



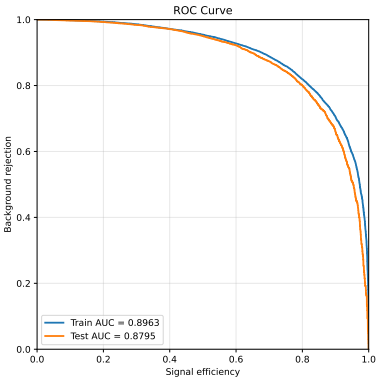
1000 fixed training rounds; no early-stopping optimum.

ROC curve: Btrk1dR, Btrk2Pt, Btrk1Pt

2018

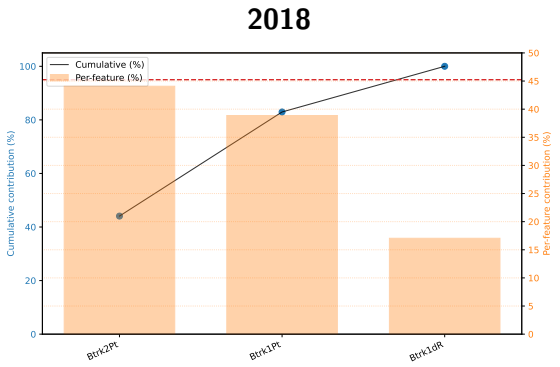


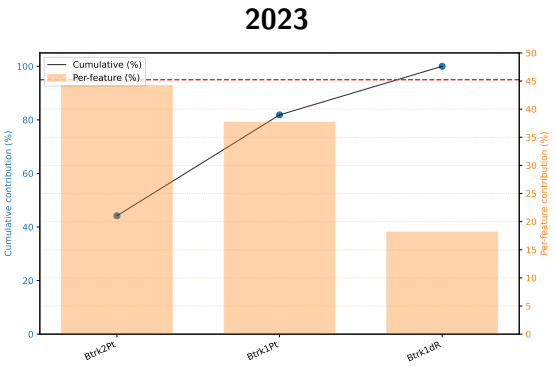
2023



Signal: physical MC weights; background: unit weights. Train/test and pooled curves retain their original definitions.

Cumulative SHAP: Btrk1dR, Btrk2Pt, Btrk1Pt

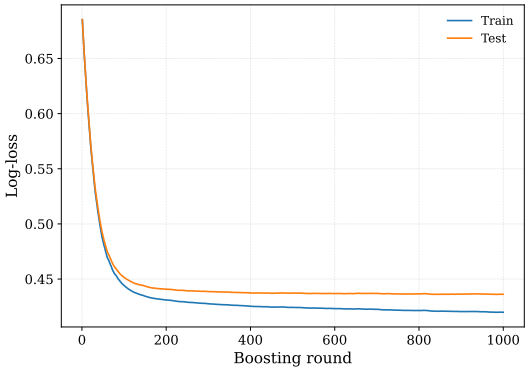




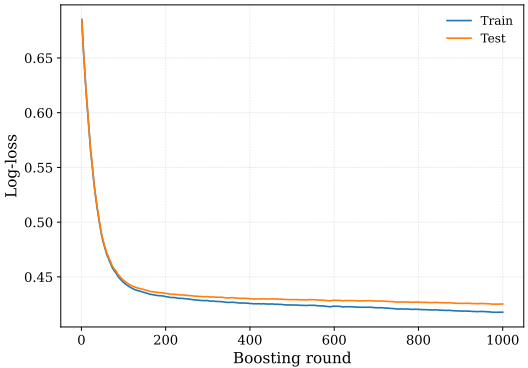
Cumulative weighted mean $|\mathrm{SHAP}|$ on a mixed signal/background sample.

Logloss: BtrkLeadPt, BtrkSubPt, BtrkMaxdR

2018



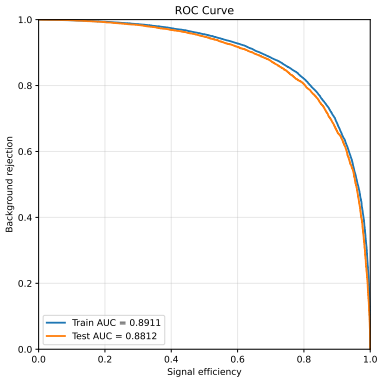
2023



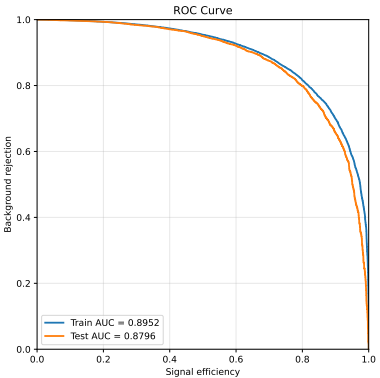
1000 fixed training rounds; no early-stopping optimum.

ROC curve: BtrkLeadPt, BtrkSubPt, BtrkMaxdR

2018

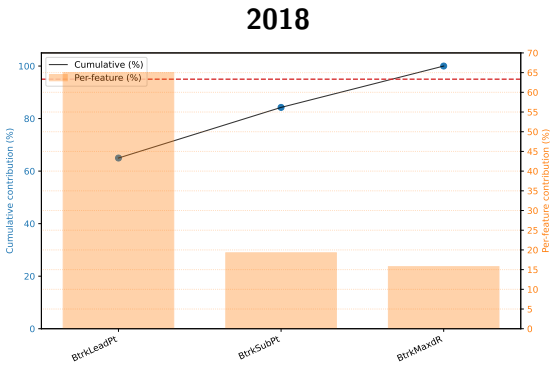


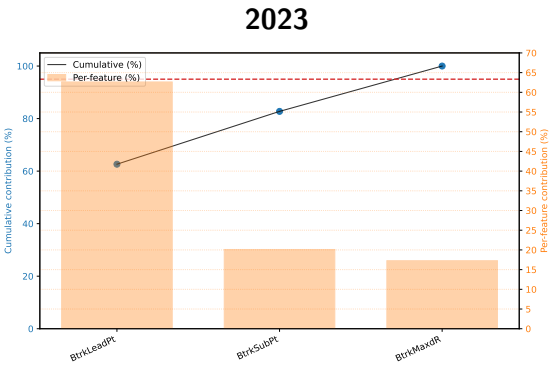
2023



Signal: physical MC weights; background: unit weights. Train/test and pooled curves retain their original definitions.

Cumulative SHAP: BtrkLeadPt, BtrkSubPt, BtrkMaxdR



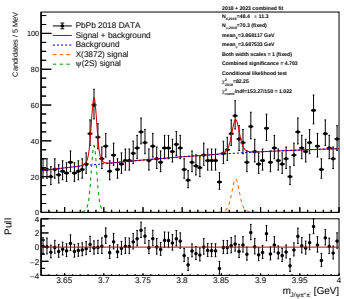


Cumulative weighted mean $|\mathrm{SHAP}|$ on a mixed signal/background sample.

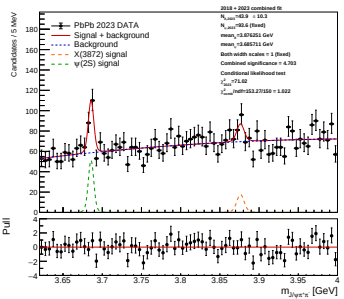
9v1: combined 2018 + 2023 fit

9v1: `Bmu1pt`, `Bmu1eta`, `Bmu2pt`, `Bmu2eta`, `Btrk2Pt`, `BtrkPtimb`, `Btrk1dR`, `Bujeta`, `Btktketa`.

2018 score > 0.88



2023 score > 0.85

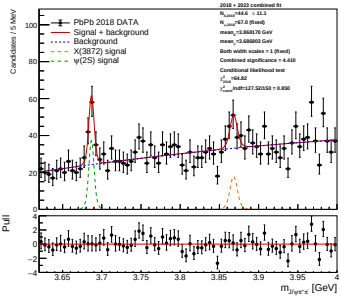


Combined: $N_X = 92.27 \pm 21.52$; significance $= 4.703\sigma$.
Independent-year reference: 2018: 3.903σ; 2023: 2.587σ.
Combined: $\chi^2/\mathrm{ndf} = 1.022$.
MC double-Gaussian widths fixed; 2nd-order Chebyshev background.

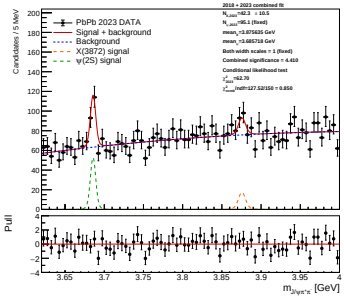
5v1: combined 2018 + 2023 fit

5v1: `Bmu1pt`, `Bmu2pt`, `Btrk2Pt`, `BtrkPtimb`, `Btrk1dR`.

2018 score > 0.89



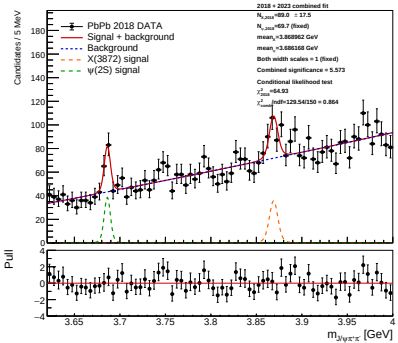
2023 score > 0.85



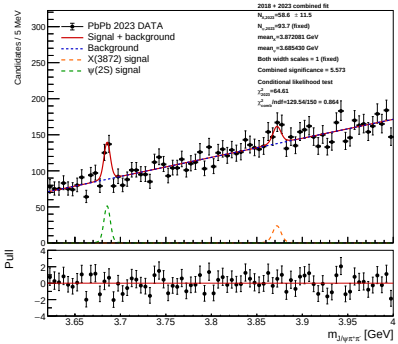
Combined: $N_X = 86.96 \pm 21.56$; significance $= 4.410\sigma$.
Independent-year reference: 2018: 3.671σ; 2023: 2.397σ.
Combined: $\chi^2/\mathrm{ndf} = 0.850$.
MC double-Gaussian widths fixed; 2nd-order Chebyshev background.

Combined fit: Btrk1dR, Btrk2Pt, BtrkPtimb

2018 score > 0.83



2023 score > 0.80



Combined: $N_X = 147.69 \pm 28.94$; significance $= 5.573\sigma$.

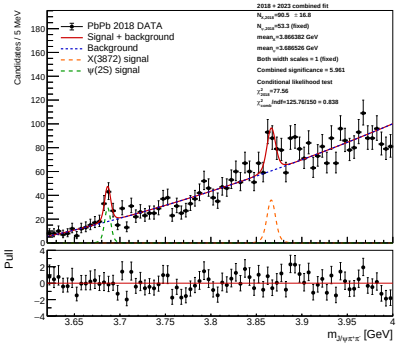
Independent-year reference: 2018: 4.949σ; 2023: 2.466σ.

Combined: $\chi^2/\mathrm{ndf} = 0.864$.

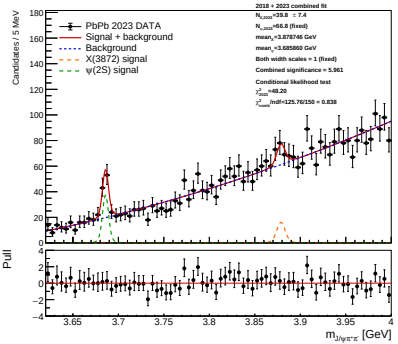
MC double-Gaussian widths fixed; 2nd-order Chebyshev background.

Combined fit: BtrkLeadPt, BtrkPtimb, BtrkMaxdR

2018 score > 0.85



2023 score > 0.88



Combined: $N_X = 130.38 \pm 24.21$; significance $= 5.961\sigma$.

Independent-year reference: 2018: 5.400σ; 2023: 2.460σ.

Combined: $\chi^2/\mathrm{ndf} = 0.838$.

MC double-Gaussian widths fixed; 2nd-order Chebyshev background.